



iKnowiSee: AR Glasses with Language Learning Translation System and Identity Recognition System Built Based on Large Pre-trained Models of Language and Vision and Internet of Things Technology

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Abstract. AR glasses used in daily life have made good progress and have some practical value. However, the current design concept of AR glasses is basically to simply port the content of a cell phone and act as a secondary screen for the phone. In contrast, the AR glasses we designed are based on actual situations, focus on real-world interactions, and utilize IoT technology with the aim of enabling users to fully extract and utilize the digital information in their lives. We have created two innovative features, one is a language learning translation system for users to learn foreign languages, which integrates a large language model with an open vocabulary recognition model to fully extract the visual semantic information of the scene; and the other is a social conferencing system, which utilizes the IoT cloud, pipe, edge, and end development to reduce the cost of communication and improve the efficiency of exchanges in social situations.

Keywords: LLM · Open Vocabulary Object Detection · IoT · Augmented reality · Wearable device

1 Introduction

At present, with the updating and progress of integration, display, human-computer interaction, augmented reality and other technologies, the development of AR glasses has reached a more mature stage, and has a wide range of applications in daily life, audio-visual entertainment, industry, medical and other fields [1, 2]. At present, most of the AR glasses on the market for daily life use are transplanting the content displayed on the cell phone to the glasses, acting as the screen projector of the cell phone, aiming at replacing the smartphone with the carrier of the glasses. However, this approach does not start from the basic nature of the glasses, and its design starting point does not start from the real-life scenarios, but simply moves the cell phone screen to the glasses, which is disconnected from the real life and produces a certain sense of separation, and does not make full use of the maximum value of the glasses as a wearable device in reality.

Instead, we have designed an innovative concept for AR glasses that starts from context and scene. We design AR glasses from the actual application scenarios, with the actual needs of the scene as the core, based on the Internet of Things cloud, pipe, edge, end, design and use of adapted algorithms to match, through the slice of the life situation for the operation, build iKnowiSee exclusive APP to integrate all functions, so that the virtual information is more naturally integrated into the real life, and to realize the augmentation of both augmentation and reality. At present, we have built and realized many daily functions on AR glasses, which mainly include two innovative functions, i.e. identification system and language learning and translation system, which meet the needs of people's daily socialization and foreign language learning, and effectively validate the feasibility of the design concept of our AR glasses.

Our language learning translation system makes full use of the visual information of real-life scenes, which can capture the information of the scenes seen by the user and convert it into corresponding translation content, phonetic symbols and example sentences. We utilize the Big Language Model and the Open Lexical Identification Model CLIP [3] to complete the content conversion from real-world information to multi-language translations, and propose a new type of language learning through wearable devices, i.e., the learning process from visual information to language output. Utilizing the flexibility of the large language model and the richness of the corpus, users can design the style of the example sentences generated by LLM according to their own interests and meet the learning needs of users of different languages. At present, this system supports more than fifty foreign language learning styles. Unlike the AR glasses on the market that are only limited to using voice information for simultaneous interpretation, we fully extract the semantic information contained in the visuals of real-life scenes and present it to the user.

We combined AR glasses with IoT technology to create a conference system, aiming to improve the efficiency of conference communication and reduce communication costs. We use AR glasses as a carrier, and use Harmony OS APP development, cloud technology and face recognition algorithms to combine to design personalized AR post-cards for attendees, to create an adaptable conference system, so as to achieve two-way transmission of both parties' identity information in the conference. We also migrate the conference system to design a social system, by dividing different groups into different circles, so that users can understand the information of other like-minded people, promote the communication between the two sides, and enhance the social dating atmosphere and togetherness. At the same time, we also designed a specialized privacy policy to protect user information.

In summary, the main contributions of this paper are: (1) We designed a new AR glasses design concept, that is, from the actual scenario, for the needs of real-world scenarios to provide adaptive services, and through the APP integration of all the functions, only need to be authenticated by the account can be used across the device, instead of simply transplanting the content of the phone to the AR glasses. to the AR glasses program. (2) Design a new foreign language learning system, and use the open vocabulary recognition model CLIP and the large language model to realize the conversion of scene visual information to multilingual multi-level content linguistic information, without the need to label the data set, can be generalized to real life to complete the recognition

and translation, with good robustness. (3) Combine the conference scene with AR and IoT technology to create and design personalized AR postcards for users to improve the efficiency of conference communication, and at the same time, migrate the system to expand the offline AR dating method, and move the Internet dating strategy to the offline AR display.

2 Related Work

A Multilingual and Multilevel Vision-Language Learning Paradigm. Currently, there are no AR glasses on the market that specialize in recognizing external information and converting it into foreign language information for learning [4–6]. Other smart devices are also limited to the application of knowing everything, without considering linking the recognized content to the context of language learning [7–9]. We have successfully deployed the system on AR glasses by presenting the information from interaction with the external environment in a more flexible form and multimodal paradigm. The system makes full use of the environmental visual information, performs feature extraction on the perceived external information, and reflects the inferred information about the objects to the user in the form of translated content, phonetic symbols, and example sentence demonstrations, so that the user can obtain foreign language information about the objects he/she sees. We use the open vocabulary detection model CLIP as the basis of visual perception and inference for AR glasses to achieve powerful visual generalization perception, which is more suitable for coping with the diversity and complexity of information in real-life scenes, and flexibly adapts to the demanding requirements of visual recognition language learning. We have also embedded a large language model to achieve multi-language, personalized and flexible teaching example output.

Meeting and Social Identity Recognition System Based on IoT Architecture. Most of the current AR glasses with face detection algorithms have limited application scenarios and are only suitable for some mobile people recognition scenarios, such as recognizing unfamiliar people [10–14]. This kind of AR glasses lacks the role of the scene of human-human interaction, while in the use of our AR glasses, the use of IoT technology to break the traditional single human interaction with the glasses, through the connection of things to things to realize the connection between people and people. Our AR glasses are integrated with the identity recognition system, which can obtain other people's identity information, including department, position, etc. People and people use AR glasses as a medium to realize the information interaction function of people recognizing each other. Users register basic identity information and design additional display of personalized information through APP, and the information is uploaded and stored in the cloud. Users can display each other's postcard information when they participate in the meeting and detect others, and they can still query each other's information when the distance between people is far, and pass their own AR postcards to other participants in the form of augmented reality technology. The use of identity information transfer in the form of AR improves the efficiency of information flow in the meeting and helps participants to obtain and express information.

3 Methods

Our objective is to craft intelligent augmented reality (AR) glasses that are rooted in real-world scenarios, ensuring functionality tailored to practical contexts [15–18]. Presently, we have developed two primary practical scenarios: language learning translation and social interaction during meetings. We leverage the versatility of advanced language models, diverse deep learning algorithms, and Internet of Things (IoT) development techniques to fulfill the functional requirements of these glasses.

3.1 Overall Framework and Implementation of AR Glasses Designed and Developed Based on Internet of Things Technology

We have conceptualized and manufactured an intelligent wearable device based on Internet of Things (IoT) technology, artificial intelligence, and operating system development techniques. This device facilitates audio-visual information interaction through physical manipulation and voice communication, integrated with augmented reality (AR) glasses [7, 19–21]. The system operates on a foundation of cloud, edge, and endpoint, deploying computationally intensive models on cloud servers, loading them into cloud storage, and swiftly processing digital information and multimodal data. Communication between local devices and the cloud for real-time interaction is ensured through protocols such as Socket and Https, forming the basis for seamless information exchange and data communication among the eyewear device, cloud platform, and application (APP). Some lightweight computing and data processing occur through edge computing on the AR glasses, utilizing their built-in information processing capabilities for rapid data transformation. The technical architecture, based on cloud, edge, and endpoint, integrates IoT technology into AR glasses, achieving an operational mode that unifies artificial intelligence, operating systems, and transmission technologies.

The project incorporates a wearable peripheral ring, utilizing 2.4G transmission, to remotely control the AR glasses, enhancing functionality through an easily interactive interface. The AR glasses feature a compact, high-resolution projection screen that can connect to various chip styles to meet diverse practical requirements, ensuring excellent scalability. External edge devices are employed to accomplish complete image decoding, transmission, and display. The precise structure of optical prisms enables real-time reception of decoded images from the OLED display, providing clear and high-quality visuals to the eyes while allowing users to observe external natural information unhindered, significantly reducing the dissonance between AR screen displays and real-life surroundings.

The AR glasses are controlled either through the ring or voice interaction, coupled with perceived external information. Encoded information is transmitted via protocols like Socket to the cloud (we utilize Huawei Cloud for data operations and algorithm model deployment). On the cloud, the face detection system is deployed on ECS servers, awaiting commands through the APP and AR glasses' mode selection for information input or personnel recognition. Relevant personnel information is stored in OBS data buckets for collective coordination and management. The visual perception and recognition algorithm deployed on Model Arts integrates with LLM for multimodal integration, transforming the voice and visual information transmitted from the AR glasses into the

desired language information through model extraction and inference, returning it to the AR glasses through the transmission protocol.

The APP, developed on Harmony OS (Hongmeng operating system), serves as a dedicated companion application for input and output information, acting as a bridge between users and AR glasses. It facilitates information input, mode switching, data viewing, scene storage, and real-time communication with the cloud, acting as an interactive intermediary between users and the cloud computing center (Fig. 1).

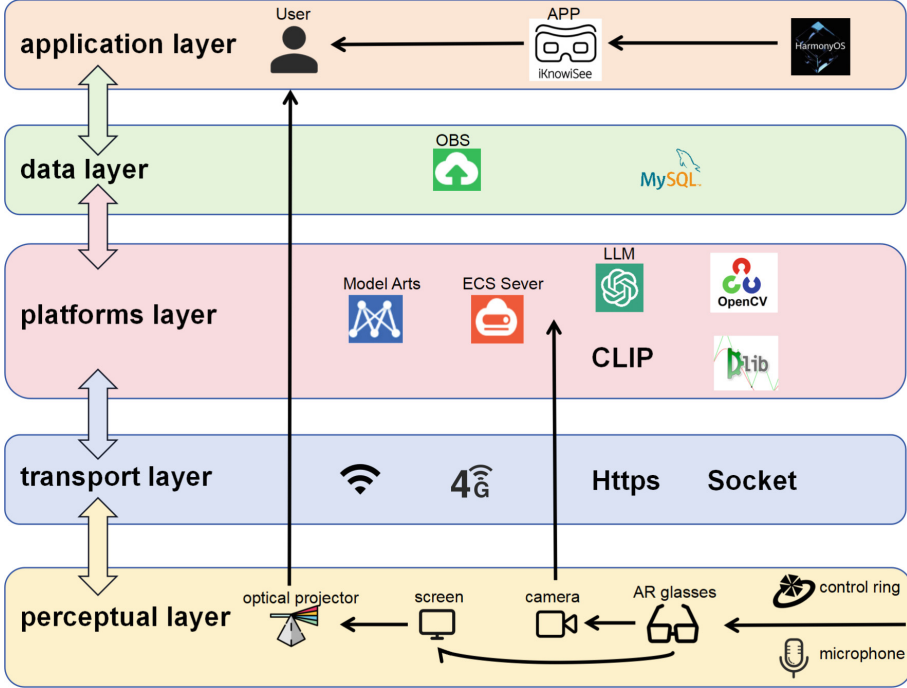


Fig. 1. System architecture diagram based on IoT hierarchy

3.2 Language Learning and Translation System

The AR glasses perceive overall scene information and, triggered by user ring manipulation or voice activation, initiate the visual detection system. The detected scene content is transmitted to the CLIP visual language model for recognition [22–25]. CLIP calculates distances and matches the detected results based on corpus information and visual content. The user, in advance, sets language learning content and completes style customization on the APP. These settings are fused with CLIP’s recognition results and input into the Large Language Model (LLM). LLM, based on detection results and user settings, transforms semantic information, outputting translated content, phonetic symbols, and example sentences corresponding to the scene information, displayed on the

AR glasses. The transformed information can also be selected and stored in the APP for future retrieval and review of language-scene matching information.

Text-to-Graph Matching Function Based on Open Lexical Identification Model

CLIP. We validated the feasibility of our system using the fundamental open vocabulary recognition model, CLIP. The model automatically matches language and image information based on object scene information and a pre-set corpus, generating corresponding results. We deployed the CLIP model on the cloud, relieving local processing pressure on the AR glasses. Leveraging the Huawei Cloud Model Arts platform as an intermediary and utilizing cloud platform GPU computing resources, CLIP object recognition is accelerated, ensuring the timeliness of the language learning system.

Multi-language Content Conversion Based on Large Language Model. Large language models exhibit randomness in generating content, aligning well with our system's flexibility requirements for foreign language teaching. The same scene can yield diverse and multilingual outputs. By combining CLIP with LLM, visual detection information is transformed into prompt forms, integrating user-provided language and style customization information into pre-built LLM prompt templates, achieving personalized output. The real-time changeability of scene information, language selection, and style input provides ample flexibility for the language learning system.

Text Translation Combining Large Language Model and OCR Technology. LLM's application extends to additional functions in our system. AR glasses can extract text information from the scene, and through cloud-deployed OCR algorithms, exact text is extracted. Based on the user's native language environment, the text information is filled into the LLM prompt template, facilitating the conversion of external text content into native language information for quick foreign language reading by the user (Fig. 2).

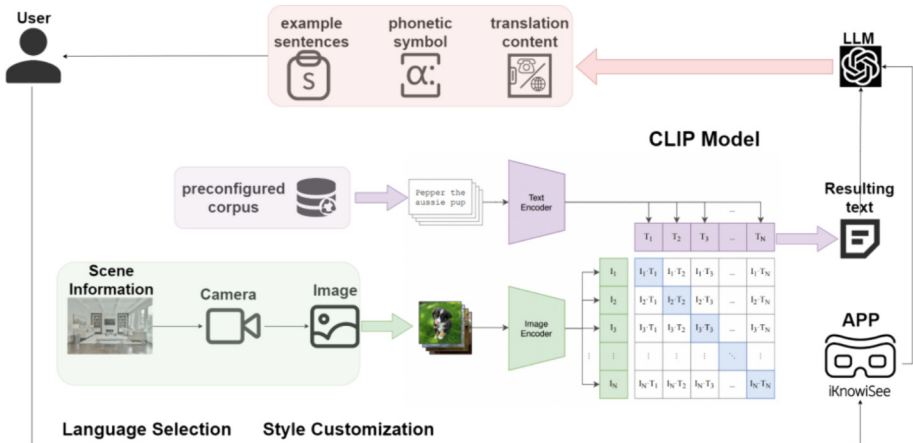


Fig. 2. Language Learning and Translation System Overall Flowchart

3.3 Identification System

During the identity information entry process, users register their identity information through the APP, complete personalized content, and input facial information. The APP collects facial and text data, transmitting it to the cloud server [26–30]. The face detection algorithm deployed on the ECS server extracts the user's facial feature vector and writes it into a CSV file. The file content and user text information are then entered into the OBS database for future reference. When users invoke the recognition function to detect other members' faces, the algorithm is activated through ring manipulation or voice command. The camera captures facial data, uploads it to the cloud, extracts facial feature vectors, matches the detected facial data with the CSV file in the database, calculates the minimum Euclidean distance, and finally matches the object's identity information, displaying it on the AR glasses.

IoT Identification System Combining AR Glasses Entity, Face Detection Algorithm, APP, and Cloud Technology. The use of the algorithm involves two stages: the identity information entry stage and the recognition stage. The face detection algorithm combines OpenCV and DLIB libraries, utilizing Resnet to extract facial features. It integrates facial keypoint information obtained by DLIB to obtain facial feature vectors, which are then written into a CSV file and uploaded to the cloud server. In the identity entry stage, users use the APP for identity entry, and the APP transmits the collected facial information and corresponding data to the cloud for record storage. In the recognition stage, the AR glasses capture the members' faces, encode the visual information, upload it to the cloud for face detection, decode the recognition results, and transmit them back to the AR glasses for display to the user (Fig. 3).

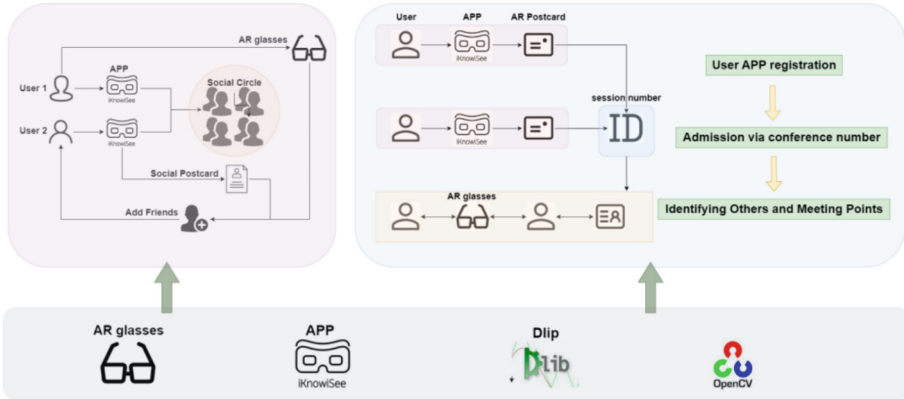


Fig. 3. Overall framework of the identification system

4 Experiments

We deployed relevant models and algorithms on the fabricated AR glasses to demonstrate the actual effects of the language learning translation system and the social conferencing system respectively.

4.1 AR Glasses Entity

Figure 4 shows the overall structure of the AR glasses hardware on which we deployed the model and through which we tested the functionality.

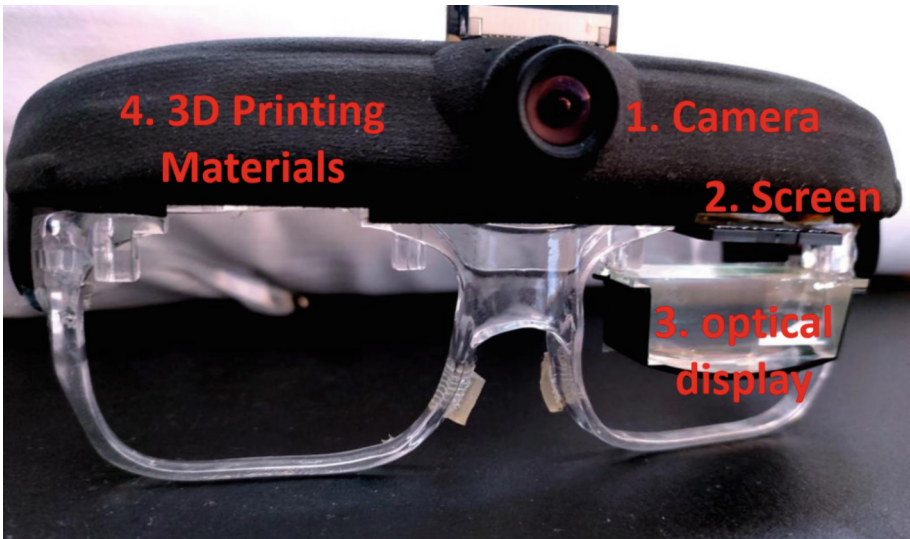


Fig. 4. AR glasses physical picture

4.2 Language Learning and Translation System

As shown in Figs. 5 and 6, the first-person perspective capture image and the AR glasses screen display image are shown respectively. The system takes the image capturing device mounted on the AR glasses as data input, and the visual language model CLIP deployed to the cloud parses the transmitted data, converts the parsed content into the final result by using the large language model, and outputs the result as translated content, phonetic symbols and example sentences.

The system can realize the interconversion of more than fifty languages due to the richness of the large language model corpus, and we demonstrate the scenarios of French and Japanese learning respectively in the system.

We validate the pedagogical capabilities of the language learning system paired with the open-world recognition model CLIP and the large language model by demonstrating scenarios in French and Japanese.

4.3 Identification System

As shown in Fig. 7, the third-person view graph, the first-person view graph, and the screen information graph displayed by the AR glasses seen by the user in the actual meeting and socializing occasions are shown respectively. In meetings and social occasions, the user can obtain the identity information of the other party by using the AR glasses at a long distance and display it through the AR glasses.

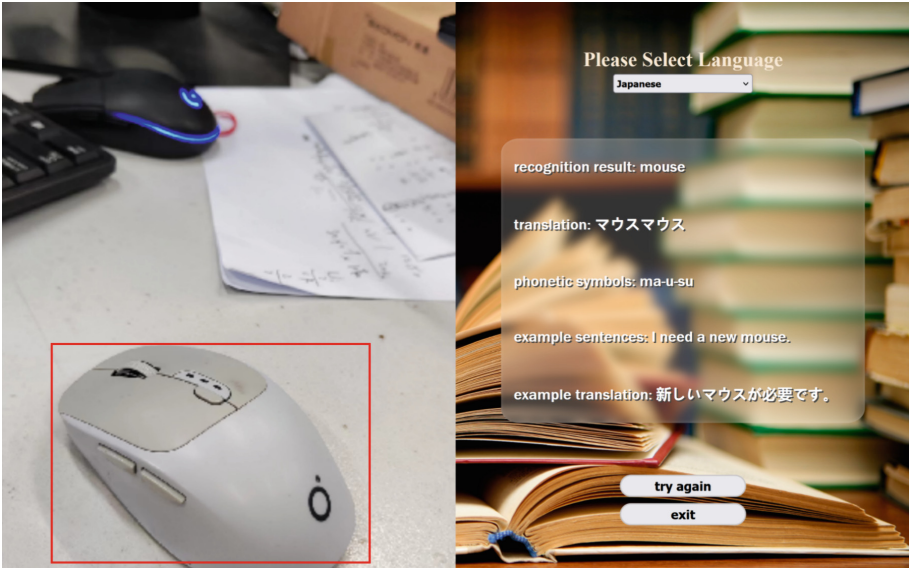


Fig. 5. The French learning scenario.

We utilize IoT technology combined with face recognition algorithm and APP development, and verified the recognition effect of our system by comparing and displaying the effect of actual scenes.

4.4 APP Design and Build

As shown in Fig. 8, some of the function pages on the APP are shown respectively, including information registration, face recognition, meeting and making friends.

The above is the APP part of the page display effect, which verifies the feasibility of our APP construction and design.



Fig. 6. The Japanese learning scenario.

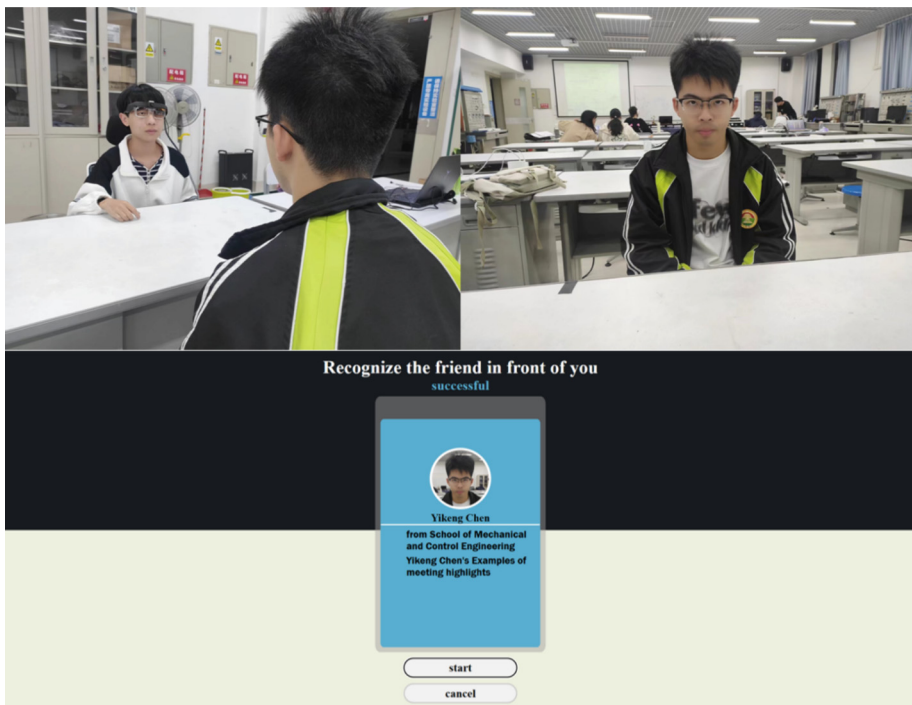


Fig. 7. Demonstration of the Identification System

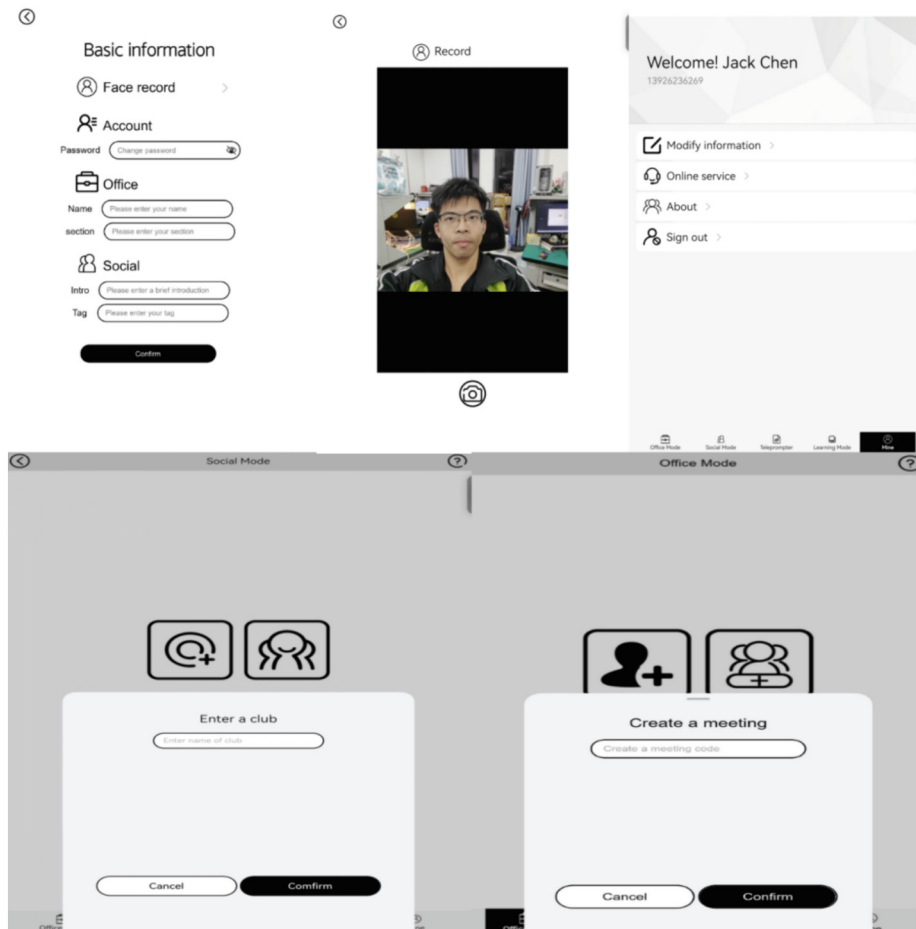


Fig. 8. APP Demo

5 Conclusion

In order to improve the multi-level learning of foreign languages, meet the learning needs of users of different languages, and improve the efficiency of communication among people in social occasions, we designed a language learning translation system and a social meeting system based on the Internet of Things architecture of cloud, management, edge and end, combined with a visual language model, a large language model, a face recognition algorithm, and an APP, and designed two experiments of real-world situation recognition and foreign language conversion and identification of people in social occasions to verify the effectiveness of this system and identity information recognition. We also designed two experiments of real situation recognition and foreign language conversion and identity information recognition of people in social situations to verify the effectiveness of the system in foreign language learning as well as identity information recognition. We deployed the two systems on AR glasses to realize the design

concept of solving specific needs based on actual scenarios from realistic situations, and we also made our own APP to integrate all the functions, and tested the self-designed software program through multiple mobile devices, which verified the good migratory capability of the designed AR glasses system framework.

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